Leveraging Verifier-Based Reinforcement Learning in Image Editing

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Research Questions

- How can we build robust reward models for image editing that provide principled, fine-grained evaluation rather than holistic scoring?
- How can we align pointwise reasoning reward models with pairwise human preference data for image editing tasks?
- How can we leverage non-differentiable reasoning-based reward models to improve downstream image editing models via reinforcement learning?

Core Idea

Existing image editing reward models give single holistic scores, failing to evaluate complex aspects like instruction adherence vs. preservation of unedited regions. This paper introduces a verifier-based approach that decomposes editing instructions into specific principles (what to keep, what to change, quality checks) and evaluates each separately using chain-of-thought reasoning. The key insight is moving from a 'scorer' to a 'verifier' - instead of outputting one number, the model reasons through each principle step-by-step before aggregating into a final score. They train this via supervised fine-tuning on high-quality reasoning traces, then use a novel GCPO algorithm to align with human preferences by contrasting groups of winner/loser reasoning trajectories.

Key Formula

GCPO Win/Loss Ratios

$$r_j^w = \frac{1}{N}\sum_{k=1}^{N} 1\{\tau_j^w > \tau_k^l\}, \quad r_j^l = \frac{1}{N}\sum_{k=1}^{N} 1\{\tau_j^l < \tau_k^w\}$$

Computes win ratio for preferred candidate τ^w_j (fraction of non-preferred candidates it beats) and loss ratio for non-preferred candidate τ^l_j (fraction of preferred candidates that beat it). N is number of reasoning traces, indicator function 1{·} counts comparisons. Used to convert pairwise preferences into rewards for pointwise model training.

GRPO Advantage Computation

$$A_i = \frac{\tau_i - \mathrm{mean}\{\tau_i\}_{i=1}^{G}}{\mathrm{std}\{\tau_i\}_{i=1}^{G} + \epsilon_{std}}, \quad \tau_i = \Phi(R_\phi(x_0^i, c, P))$$

Computes advantage A_i for i-th generated image in group of G samples. τ_i is scalar reward from RRM R_φ extracted by operator Φ, c is context, P is principles. Advantage normalizes reward against group mean/std for stable GRPO training of editing models.

Key Figure

3.1.2 Reasoning-Reinforced Reward Learning

Although the reward model possesses effective Chain-of-Thought (CoT) reasoning capabilities following the cold-start SFT phase, we observe that its judgments can be fallible. The model may exhibit hallucinations or struggle to accurately assess the magnitude of edits, such as incorrectly verifying a principle "move to the left of the figure" as successful, but the object has only slightly moved, as detailed in the Appendix.

To further align the model with human preferences, we introduce a reinforcement learning phase. A key challenge is that the RRM first generates a reasoning trace before producing a final score, making it difficult to optimize with standard scalar-reward formulations directly.

As illustrated in Fig. 2, this phase employs an inter-reward, intra-advantage based GRPO algorithm, which we term Group Contrastive Preference Optimization (GCPO), to refine the reward model using human-annotated preference pairs. In this framework, the reward model R_ϕ itself serves as the policy being optimized, where ϕ are its parameters. The "actions" consist of the generated reasoning trace and the final score, which are produced conditioned on the input quadruple data.

Preference Data. For this phase, we construct a preference dataset, $\mathcal{D}$, through human annotation. The annotation process was as follows: annotators were presented with a source image x_{ref}, an editing instruction q, and a pair of edited images. They were asked to choose which image was better, or to label them as "same" if they were of comparable quality or if a clear preference could not be established. The primary criteria for judgment were instruction fidelity and overall image quality. This process yielded a dataset of approximately 10,000 preference pairs (x^w, x^l) for each context $c = (x_{\text{ref}}, q)$, where x^w denotes the preferred (winner) image and x^l the non-preferred (loser) one. The "same" pairs were excluded from the GCPO training.

Win/Loss Ratio Rewards. We employ pairwise win/loss ratio rewards derived from cross-group preference comparisons. For each preference pair (x^w, x^l), the reward model $\mathbb{R}_\phi$ stochastically generates N distinct reasoning traces and their corresponding scores $\{\tau_j^w\}_{j=1}^N$ and $\{\tau_j^l\}_{j=1}^N$ for each image:

$$\tau_j^w = \Phi\big(\mathbb{R}_\phi(x_j^w, c, \mathcal{P})\big)\,, \quad \tau_j^l = \Phi\big(\mathbb{R}_\phi(x_j^l, c, \mathcal{P})\big)\,, \tag{1}$$

where $\Phi(\cdot)$ is an operator that extracts the scalar score from the text output of $\mathbb{R}_\phi(\cdot,\cdot,\cdot)$ via rule-based parsing.

The per-sample win/loss ratios are then defined based on exhaustive pairwise comparisons between the two sets of scores, ignoring ties. The **win ratio** for a preferred candidate τ_g^j is the fraction of non-preferred candidates it scores higher than. Symmetrically, the **loss ratio** for a non-preferred candidate τ_b^j is the fraction of preferred candidates that score lower than it:

$$r_j^w = \frac{1}{N}\sum_{k=1}^{N} \mathbb{1}\{\tau_j^w > \tau_k^l\}\,, \quad r_j^l = \frac{1}{N}\sum_{k=1}^{N} \mathbb{1}\{\tau_j^l < \tau_k^w\}\,, \tag{2}$$

where N denotes the number of reasoning traces generated, $\mathbb{1}\{\cdot\}$ is the indicator function.

Optimization with GCPO. After computing the win/loss ratio rewards $\{r_j^w\}_{j=1}^N$ and $\{r_j^l\}_{j=1}^N$ from cross-group comparisons, the original pairing between samples is disregarded for the optimization step. Instead, advantages are computed independently within each rollout group (preferred or non-preferred). Although the rewards originate from paired comparisons, the loss is calculated by partitioning the rollouts into two distinct sets. The advantages are computed as follows:

$$\begin{aligned} \bar{r}^w &= \frac{1}{N}\sum_{j=1}^{N} r_j^w, \qquad \bar{r}^l = \frac{1}{N}\sum_{j=1}^{N} r_j^l, \\ A_j^w &= r_j^w - \bar{r}^w, \qquad A_j^l = r_j^l - \bar{r}^l. \end{aligned} \tag{3}$$

Let $r_{t,j}^w(\phi)$ and $r_{t,j}^l(\phi)$ denote the per-token likelihood ratios for the j-th rollout and t-th token in the preferred and non-preferred groups, respectively. The objective function is the sum of the two groups' clipped surrogate losses, omitting the KL divergence term:

Figure 2: Shows the complete two-stage RRM training pipeline including cold-start SFT with principle decomposition and VLM verification, plus GCPO with cross-group win/loss ratio computation.

Limitations

- Requires expensive external VLM verification during SFT data construction, limiting scalability
- GCPO training needs multiple rollouts from both preferred and non-preferred groups, increasing computational cost
- Non-differentiable nature of RRM limits integration with gradient-based optimization methods
- Performance depends heavily on quality of principle decomposition, which may not generalize to all editing tasks